

ARC Figures

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```
library(tidyverse)
library(dplyr)
library(readxl)
library(janitor)
library(ggplot2)
library(ggtext)
library(stringr)
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```
frequencies_renamed <- read_excel("Frequencies PI and Cxn to CPS - Updated Names.xlsx")
frequencies_renamed <- clean_names(frequencies_renamed) |>
  mutate(denominator = case_match(denominator,
                                   "imprisoned people" ~ "Imprisoned People",
                                   "incarcerated parents" ~ "Incarcerated Parents",
                                   "foster care children" ~ "Foster Care Children",
                                   "CPS-involved children" ~ "CPS Involved-Children"
                                   ))
```

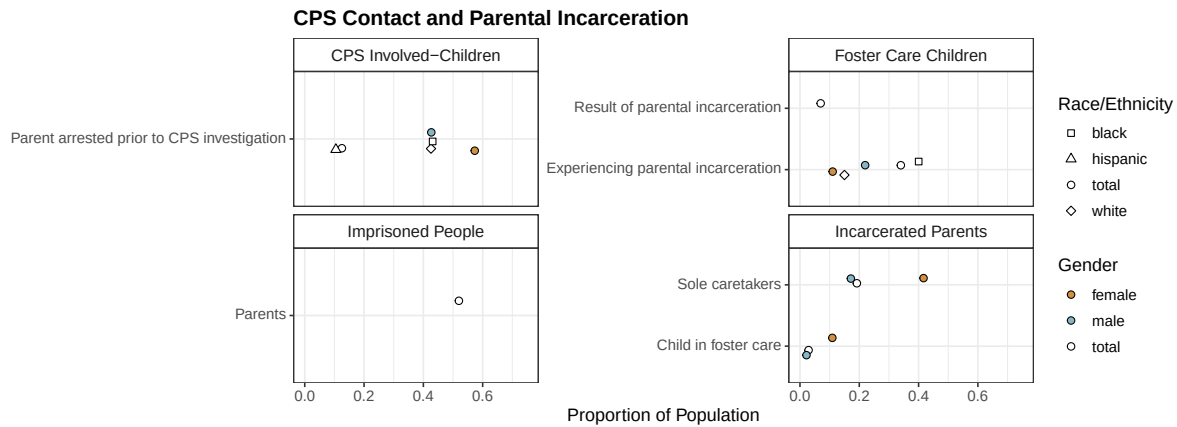
```
frequencies_pivoted <- frequencies_renamed |>
  pivot_longer(cols = c(total, male, female, white, black, hispanic),
               names_to = "demographic",
               values_to = "proportion") |>
  filter(!is.na(proportion)) |>
  mutate(race = case_when(
    demographic %in% c("white", "black", "hispanic") ~ demographic,
    TRUE ~ "total"
  ),
  gender = case_when(demographic %in% c("male", "female") ~ demographic,
    TRUE ~ "total"),
  cps_cj = if_else(denominator %in% c("Incarcerated Parents",
                                       "Imprisoned People"),
    "Criminal Justice Contact",
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                                "Child Protective Services Contact")) |>
  select(-demographic)

frequencies_pivoted |>
  ggplot(aes(x = proportion, y = clean_name,
            fill = gender, shape = race)) +
  geom_point(size = 2,
            position = position_jitter(height = .15),
            color = "black",
            show.legend = TRUE) +
  facet_wrap(~denominator, scales = "free_y") +
  scale_fill_manual(values = c("male" = "#86B6C3", "female" = "#D29141",
                              "total" = "white")) +
  scale_shape_manual(values = c("black" = 22, "hispanic" = 24,
                              "white" = 23, "total" = 21)) +
  theme_bw() +
  theme(strip.background = element_rect(fill = NA),
        axis.title.y = element_blank(),
        strip.text.x = element_text(
          face = "plain",
          size = 11,
          margin = margin(t = 6, b = 6)
        ),
        plot.title = element_text(face = "bold"),
        text = element_text(size = 12),
        axis.text.y = element_markdown(),
        plot.caption.position = "plot",
        plot.caption = element_markdown(hjust = 0)) +
  guides(fill = guide_legend(
    override.aes = list(shape = 21, color = "black"))) +
  xlim(0, 0.75) +
  labs(title = "CPS Contact and Parental Incarceration",
       x = "Proportion of Population",
       y = "Condition",
       shape = "Race/Ethnicity",
       fill = "Gender")

```



```
frequencies_mutated <- frequencies_renamed |>
  pivot_longer(cols = c(total, male, female, white, black, hispanic),
    names_to = "demographic",
    values_to = "proportion") |>
  filter(!is.na(proportion)) |>
  mutate(race = case_when(
    demographic %in% c("white", "black", "hispanic") ~ demographic,
    TRUE ~ "total"),
  gender = case_when(demographic %in% c("male", "female") ~ demographic,
    TRUE ~ "total"),
  alt_denominator = case_when(
    str_detect(clean_name, "parental incarceration") ~ "Foster care and CPS-involved children",
    str_detect(clean_name, "Parental incarceration") ~ "Foster care and CPS-involved children",
    str_detect(clean_name, "Parent arrested") ~ "Foster care and CPS-involved children",
    TRUE ~ "Incarcerated people"),
  ) |>
  select(-demographic)

frequencies_mutated |>
  ggplot(aes(x = proportion, y = clean_name,
    fill = gender, shape = race)) +
  geom_point(size = 2,
    position = position_jitter(height = .15),
    color = "black",
    show.legend = TRUE) +
  facet_wrap(~alt_denominator, scales = "free_y") +
  scale_fill_manual(values = c("male" = "#86B6C3", "female" = "#D29141",
    "total" = "white")) +
  scale_shape_manual(values = c("black" = 22, "hispanic" = 24,
```

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"white" = 23, "total" = 21)) +
theme_bw() +
theme(strip.background = element_rect(fill = NA),
      axis.title.y = element_blank(),
      strip.text.x = element_text(
        face = "plain",
        size = 11,
        margin = margin(t = 6, b = 6)
      ),
      plot.title = element_text(face = "bold"),
      text = element_text(size = 14),
      axis.text.y = element_markdown(),
      plot.caption.position = "plot",
      plot.caption = element_markdown(hjust = 0)) +
guides(fill = guide_legend(
  override.aes = list(shape = 21, color = "black"))) +
xlim(0, 0.75) +
labs(title = "CPS Contact and Parental Incarceration",
     x = "Proportion of Population",
     y = "Condition",
     shape = "Race/Ethnicity",
     fill = "Gender")

```

